

Maschina

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Letting software hold money, do real work, and be held to account for all of it

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Abstract

Software can now decide what to do with money competently enough to be useful. It cannot be trusted to hold money, because every system in common use places the limits on that software inside the software itself, where one defect or one stolen credential removes them all at once. Maschina moves every limit outside the executing code. A machine holds funds in a wallet whose policy is enforced by a provider that knows nothing about Maschina and permits exactly one destination, which is the owner. Spending is governed by a ledger that settles against the true cost read back from the chain, actions are checked by a service holding no key, and the work is performed by nodes holding no keys at all. Every decision and its reasoning is appended to a record that cannot be rewritten. What emerges is a unit that can be funded, instructed and left alone.

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1. Introduction

There is a gap in the middle of every conversation about software that acts on its own, and it is not a gap in capability.

Software can read a market, weigh a decision and execute faster than any person. What it cannot do is be given money. Not because it would spend it foolishly, though it might, but because the arrangement that is supposed to stop it spending everything is an arrangement it administers itself.

Look at how an automated trader is actually assembled. It holds a private key, or holds a session that can reach one. Its limits sit in its own configuration, read by its own code, enforced at its own discretion. When people say such a system is “bounded”, what they mean is that it has agreed to be. The bound and the bounded party are the same party.

That arrangement survives ordinary conditions and fails under the conditions that make bounds necessary: an unexpected market, a defect nobody found, an instruction that means something different than intended, a dependency that was replaced last Tuesday, a key copied from a laptop. And when it fails it fails completely, because there is nothing underneath it. The money leaves and the chain does not offer refunds.

The scale of this is measurable. Chainalysis attributes roughly seventeen billion dollars of losses to scams and fraud in 2025, with a further three and a half billion stolen outright [1]. Private key compromise, rather than any flaw in a protocol or a contract, accounts for around forty percent of the stolen total [2]. The dominant failure in this industry is not that the logic was wrong. It is that something held a key and should not have.

Meanwhile the rails for software to spend money are being laid at speed. The x402 payment standard put Visa, Stripe, Google, AWS, Coinbase, Circle, Mastercard and the Solana Foundation into a single foundation in April 2026 [8], and Solana now settles somewhere between a half and two thirds of all traffic on that protocol [7]. The industry is busy making it easy for software to pay for things. It has not been nearly as busy making it safe.

That gap is what this paper is about.

2. The thesis

The scarce resource in a machine economy is not intelligence, strategy or compute. All three are abundant, and cheaper every quarter. The scarce resource is delegated trust: the ability to hand money to software and then stop watching it.

Whatever provides that becomes infrastructure, because everything else has to pass through it. A strategy is worth something to the person who wrote it. A custody layer that makes any strategy safe to run is worth something to everyone who will ever write one.

So Maschina is built in an order that looks backwards from outside. Custody first, capability second. For a period that produces a system with every guarantee and very few tricks, which is an uncomfortable thing to demonstrate. It is still the correct order, because capability added to a trustworthy foundation is engineering, while trust added to a capable system is a rewrite.

3. Design principles

Six commitments govern every decision in the system. They are stated here because most of the paper is their consequences.

Limits live where the limited party cannot reach them. Anything a machine enforces about itself is documentation. Only what the wallet provider, the ledger or the rule engine enforces is a limit.

Absence is never permission. A machine with no daily cap has no permission to trade, rather than unlimited permission to trade. Every system that has ever lost money to a default disagreed with this.

The record is permanent. State is not stored and updated. It is derived from an ordered append-only sequence of events that no part of the system can rewrite, including us.

Refusal is cheap and reversal is impossible. Every check happens before the signature, because on a public chain there is no afterwards.

One destination. Money enters a machine from its owner and returns to its owner. There is no other path, and the restriction is enforced outside our software.

Nothing is trusted because it is ours. Our servers, our code and our operators are treated as potentially compromised, and the guarantees are written so that assumption costs a user nothing.

4. The machine

The unit of the system is a machine: software with a job, a wallet, a budget, a mandate and a history.

Its **wallet** is created for it at a regulated key management provider. Neither the owner nor Maschina holds the key, and export is permanently denied. Creation is ordered and verified: the wallet is created, a policy is written onto it, and that policy is read back and compared against what was requested. If it does not match no machine is created. A funded wallet with nothing holding it is a state this system cannot enter.

Its **policy** names the assets it may hold, the programs it may call, and the single address it may send funds to. The provider enforces this and has never heard of Maschina.

Its **budget** is what its owner permits it to spend. Funds are reserved before an action and settled afterwards against the true cost, read back from the chain, with fees and wrapped balances accounted for, inside one database transaction holding a lock on that machine.

Its **mandate** is what its actions are checked against: that it is running, inside its window, touching only approved assets, and sized within the per-action, daily and total caps.

Its **record** is everything it did and the reasoning behind each decision.

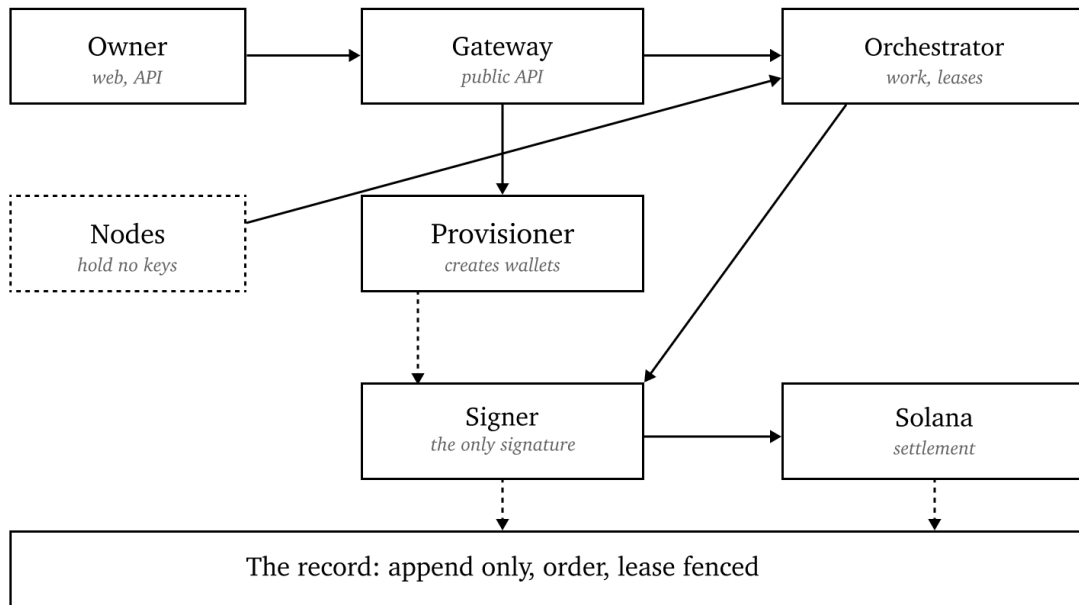


Figure 1. The service, and the boundaries enforced between them. Each boundary removes a class of failure rather than organizing code.

5. Custody, and the three refusals

Money moves only when three independent systems agree. Any one of them refuses alone.

The **mandate** is checked by a service that holds no key. The **budget** is held by a ledger under a row lock, and settled against reality rather than against an estimate. The **policy** is enforced by the provider, outside our infrastructure entirely.

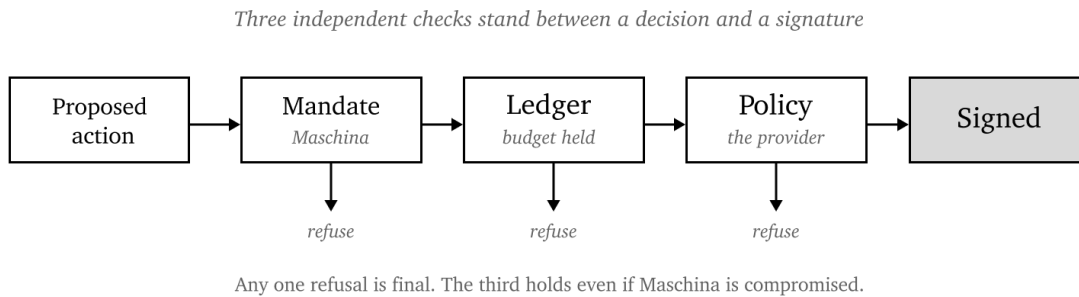


Figure 2. Three independent checks stand between a decision and a signature. The third holds even if everything Maschina operates is compromised.

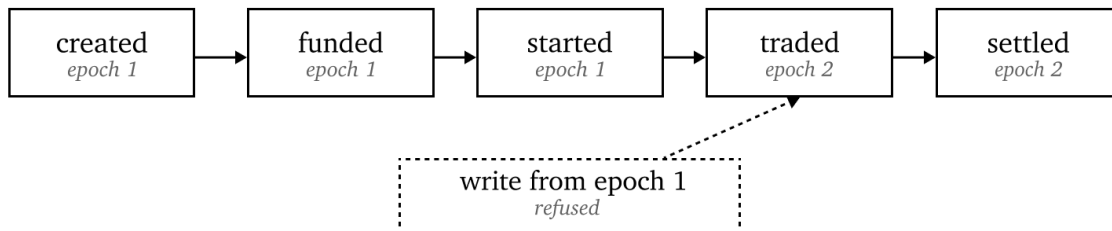
The third is the load-bearing one, and it is what makes the central claim of this paper true rather than aspirational: an attacker in complete control of Maschina’s infrastructure cannot send a user’s funds anywhere except back to that user.

Powers are separated for the same reason. The credential that creates wallets cannot sign. The credential that signs cannot create wallets or alter policies. Nodes hold no credential at all. This mirrors what the better agent wallet providers have converged on, which is a policy engine gating every signature against the principal’s rules [9], with one difference that matters: in Maschina the party being gated never holds the key in the first place.

6. The record

Everything a machine is gets derived by reading its events in order.

Two properties are enforced at the single function that writes, because anything enforced in two places is eventually enforced in none. Every event is validated against the schema for its type, since an append-only record cannot be corrected afterwards. And every write carries the lease epoch of its author, compared in the same statement that performs the insert, so a node whose work was reassigned cannot append afterwards and two nodes racing cannot both succeed.



The epoch check and the insert are one statement, so two writers cannot both win.

Figure 3. The record is append only and fenced by lease. The epoch check and the insert are a single statement, so a reassigned node cannot write history.

This is not bookkeeping. It is what makes everything in Section 9 possible, because a history that cannot be edited is a history that can be believed by someone who does not know you.

7. Execution

Work is scheduled centrally and performed by nodes. A node claims work, holds a lease and renews it while working. A node that goes quiet loses the lease and the work is reassigned.

A node reads balances, evaluates the machine’s decision, obtains a quote, checks that quote against an independent price source, and proposes. It never executes. The proposal travels to the signer, which applies the mandate, reserves budget, signs, and sends exactly once. Retries reuse the same signature rather than creating a second transaction, so a network failure at the wrong moment cannot produce a duplicate trade.

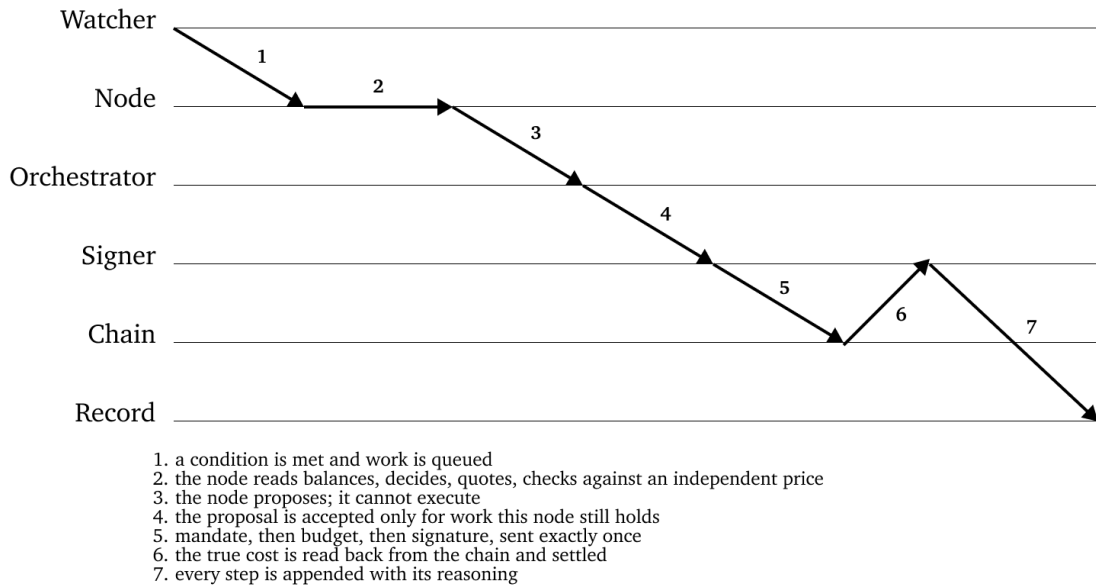
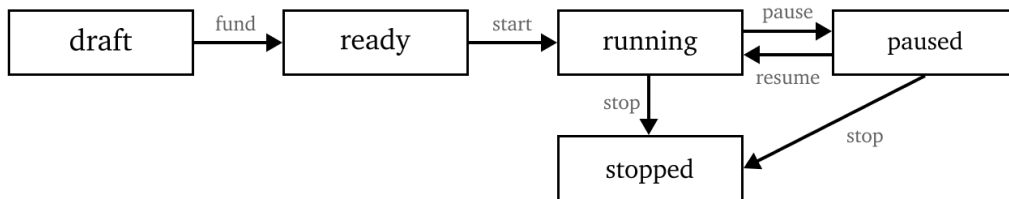


Figure 4. *One action, from condition to permanent record. The node decides and proposes; it never signs.*

Two numbers are recorded for every action: the time from decision to landing, and the difference between the quoted and the realized cost. They are the honest measure of execution quality, and they are published per machine, because a system that does not publish them is asking to be taken on faith.

8. The life of a machine

A machine moves from draft to ready when it is funded, to running when it is started, and between running and paused as its owner decides. Stopped is terminal.



Stopped is terminal. A machine that could restart itself is one you cannot switch off.

Figure 5. *The life of a machine. Stopped is final, because a machine that can restart itself is a machine its owner cannot switch off.*

That last transition is missing on purpose. An owner who stops a machine has made a decision, and a system that lets the stopped thing quietly resume has taken that decision back.

9. What is build on top

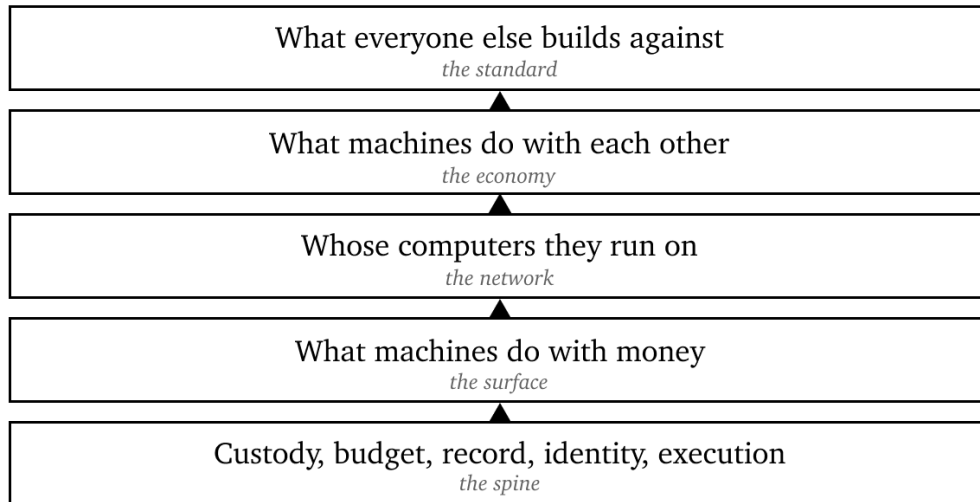


Figure 6. Each layer exists because of the one beneath it. Nothing is built that does not reuse the spine.

The surface. Trading first, because it is the hardest test of the guarantees. Then scheduled and triggered execution, positions that close themselves, ranges worked repeatedly, portfolio rebalancing, liquidity provision, lending, staking, yield on idle balances, and in time prediction markets and token issuance. Breadth is deliberate: a person who wants their money working does not want five applications, they want one place where the work happens, with one set of limits to understand.

The network. The computers that run machines belong to other people, who earn for providing them. Because nodes hold no keys, the network grows without anyone extending trust to whoever joins it. As it spreads geographically the distance between a decision and a landed transaction shortens. We do not claim to make Solana faster. We claim the shortest path from decision to settlement, and we publish the number.

The economy. Machines delegating budget under limits tighter than their own, paying for data and compute, selling what they produce, insuring each other against drawdown, and competing in public with their records as the referee.

Credit, and what follows it. Every machine accumulates a signed, verifiable history of what it did, earned, lost, and how it behaved under stress. That is a credit file. Once credit files exist, capital can be allocated to machines on evidence rather than on promises, which is a market that cannot exist anywhere the history can be edited.

10. Integrity

A system where software holds money attracts people who would like to take it, and the most common method is not sophisticated. An asset is created, bought by people who believe in it, and emptied by the person who created it. Chainalysis flagged more than sixty percent of meme coins launched in 2025 as potential rug pulls within thirty days of creation [1].

Warning people does not work. Three mechanisms do.

Mandatory asset checks, which cannot be switched off, only tightened. Authorities that permit printing, freezing or seizure. Liquidity that can be withdrawn. Ownership concentrated in a few addresses. A simulated sale against the real route, because an asset that cannot be sold in simulation is not acquired. Monitoring continues after purchase, since the events that cost people money happen after they have bought.

Standing computed from outcomes, never from opinions, because anything decided by vote is farmed within a week. Standing earns lower fees, higher limits and better placement, and losing it costs the same things. Because standing can be abandoned by starting over, the activities that matter require a bond that cannot be.

Issuance in which the issuer is paid last. Assets issued through Maschina are minted without the authorities that permit printing, freezing or seizure, with liquidity owned by the program rather than the issuer. The issuer's own allocation is held in custody by the program, unlocks against time and milestones, and on an early exit returns their cost basis and nothing more. The remainder is added to the reserve, which raises the floor for everyone still holding.

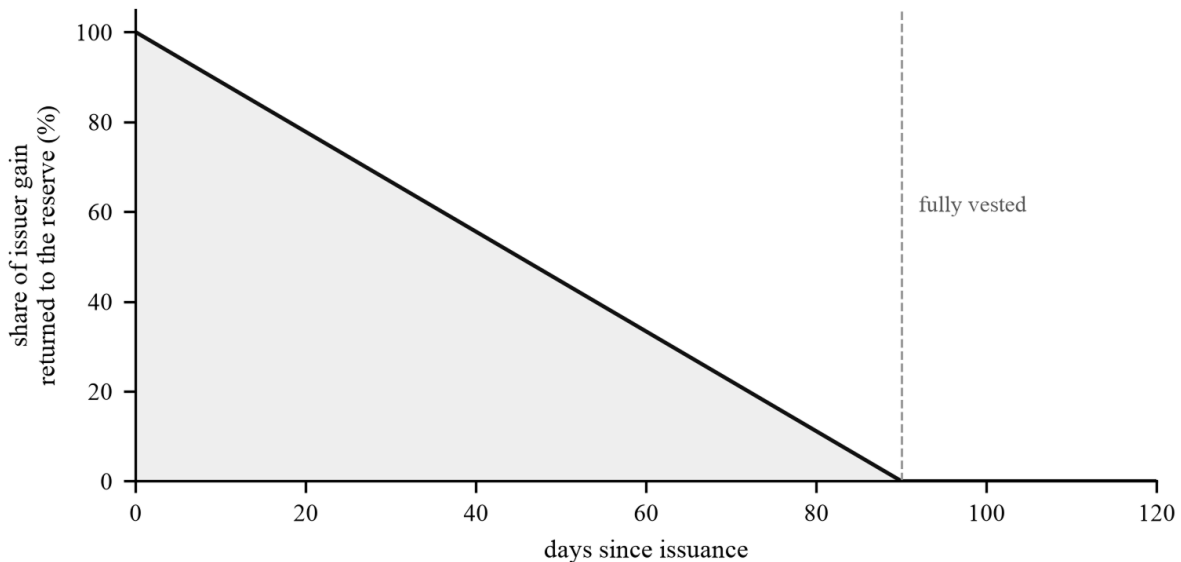


Figure 7. The share of an issuer's gain returned to the reserve, by days since issuance. The issuer is not prevented from earning. The issuer is paid last.

Two honest limits. An issuer can distribute holdings across addresses before issuance and sell from those; addresses funded by the issuer at or before issuance inherit the same rules, and concentration is detectable, but no design closes this completely. And an issuer who can never realize gain will not issue at all, which is why the confiscation decays rather than standing as a wall.

11. Economics

Maschina earns from work performed, never from outcomes promised.

Fees apply to execution, with faster paths priced above scheduled ones. A share of the fee on a copied machine goes to its creator. Subscriptions buy higher limits and a faster lane. In time, the network takes a share of what it sells and payments between machines carry a fee.

The distinction matters more than its sounds. No strategy is guaranteed to work. The academic record on retail trading is brutal and consistent: of nearly twenty thousand Brazilian day traders studied over four years, ninety seven percent of those who persisted lost money net of fees, and fewer than one in a hundred earned more than a bank teller [3]. Barber and Odean found the most active traders underperforming the market by six and a half points a year, and a fifteen year study of Taiwanese day traders found under one percent reliably profitable [4].

A business that only earns when its users profit will eventually need its users to believe they are profiting. That pressure produces every dishonest practice this industry is known for. Maschina charges for work, so it never acquires that incentive, and every result is displayed beside the benchmark of having done nothing at all.

There is no token. If one is ever issued it will buy compute, priority and fee discounts rather than represent a share of profits, and it will follow real usage rather than precede it. Maschina will not operate its own chain.

12. Implementation status

This section exists so that nothing else in this paper can be mistaken for a claim about today.

Operational as of September 20, 2026: machine wallets created with verified policies; the budget ledger, proved under concurrent load against a real database; mandate checks before signing; signing, single-send and settlement at true cost read back from the chain; quotes checked against an independent price source; the node system with leases, claims and recovery; condition monitoring that queues work; the append-only record with lease fencing; and the public API.

Under construction: machine kinds that hold and close positions, wallet sign in, the application, simulation, the marketplace, the contributed network, mandatory asset checks, standing, issuance, and every capability in Section 9 beyond immediate, scheduled and triggered execution.

Not yet performed: an independent security audit. When one occurs, its report is published in full including whatever it finds.

Not claimed: that any machine has been profitable. That claim has not been earned, and it will not be made until it can be made with the record attached.

13. Conclusion

The distance between what software can decide and what software can be trusted to hold is the whole opportunity, and it is architectural. Closing it means putting every limit somewhere the limited party cannot reach, accepting the constraints that follow, and providing the result with a history nobody can edit.

Everything else described here, the surface, the network, the economy and the markets that become possible once behavior is provable, rests on getting that one thing right first. That is the order this was built in, and it is the reason the guarantees in this paper are properties of the system rather than promises from its author.

14. References

- [1] Chainalysis. *The 2026 Crypto Crime Report*. Reports approximately \$17 billion lost to scams and fraud and \$3.4 billion stolen in 2025, an average rug pull of roughly \$510,000, and over 62% of meme coins launched in 2025 flagged as potential rug pulls within 30 days.
- [2] F. Chague, R. De-Losso and B. Giovannetti. *Day Trading for a Living?* SSRN 3423101. Of 19,646 Brazilian futures day traders studied between 2013 and 2017, 97% of those who persisted beyond 300 trading days lost money net of fees.
- [4] B. Barber and T. Odean. *Trading is Hazardous to Your Wealth*, and subsequent work on the Taiwan Stock Exchange. The most active traders returned 11.4% annually against a market return of 17.9%; under 1% of day traders were reliably profitable after fees.
- [5] A. Yakovenko. *Solana: A new architecture for a high performance blockchain*, 2017.
- [6] A. Federa, A. McConnell and M. Ward. *DoubleZero Protocol*, December 2024.
- [7] Solana Foundation and PYMNTS reporting, 2026. Since the launch of x402 on Solana, over 35 million transactions and \$10 million in volume have settled over the protocol, with Solana handling an estimated 49% to 65% of all x402 payments. Independent analysis suggest only a small fraction of that value is genuinely agent-oriented today.
- [8] The Linux Foundation. x402 Foundation announcement, April 2, 2026, with initial support including Adyen, AWS, American Express, Base, Circle, Cloudflare, Coinbase, Google, Mastercard, Microsoft, Shopify, the Solana Foundation, Stripe and Visa.
- [9] Coinbase AgentKit, Crossmint, Privy and Turnkey. Embedded wallet providers offering policy engines that gate signatures against a principal's rules.